

CHAPTER OUTLINE

ACUTE KIDNEY INJURY, 2547

SUMMARY, 2561

CHRONIC KIDNEY DISEASE, 2555

KEY POINTS

- Community-acquired AKI secondary to tropical infections, use of indigenous drugs, toxic envenomation, and obstetric factors are some of the common causes of AKI in this region.
- Peritoneal dialysis in acute settings has been used and holds promise in resource-constrained and far-off settings, where it can be used with minimal expertise and infrastructure to treat life-threatening complications of kidney failure, especially AKI.
- CKD is a major emerging noncommunicable disease in this region on account of the rapid rise in population and high prevalence of risk factors such as diabetes, hypertension, and nephrolithiasis.
- CKD of undetermined cause is common in this region and is mostly seen in young men, who are often in rural areas and work outdoors. The clinical presentation is usually late and suggests chronic tubulointerstitial involvement. It is likely that a host of environmental factors or toxins in conjunction with chronic dehydration play a role.
- Long-term renal replacement therapy is unavailable to most patients with end-stage kidney disease (ESKD) because of inaccessibility and unaffordability. However, joint public and private efforts have shown promise in establishing self-sustaining programs to provide therapy.

The Indian subcontinent occupies the southern portion of the Asian continent and includes India, Pakistan, Bangladesh, Sri Lanka, Nepal, Bhutan, and the Maldives. The subcontinent is home to well over 20% of the world's population, making it the most densely populated region in the world. These countries are all approximately at the same stage of economic development and are considered emerging and developing economies, according to the International Monetary Fund's *World Economic Outlook* report.^{1,2} Most of the population of the region lives in rural areas, relies on agriculture for its livelihood, and has limited access to health care. The countries of the subcontinent fall into the category of medium human development,³ as measured by the human development index, which takes into account not only the standard of living as determined by purchasing power parity but also the literacy rate and life expectancy.

India is the largest country in the subcontinent, with a population of more than 1 billion, and is also the largest economy in the region. It has enjoyed impressive economic growth, as a result of which India has been put into the category of "newly industrialized countries," a classification between industrialized and developing countries. The mean

per capita gross national product has grown, but 0.7 billion people still continue to live on less than one U.S. dollar a day. Economic and development indicators of major countries of the region are shown in [Table 79.1](#).^{2,3} This combination of underdevelopment, plus a rapidly growing economy, is reflected in the disease spectrum. On the one hand, countries of the subcontinent are challenged with epidemics of infectious disease and on the other hand diabetes mellitus, the so-called disease of the affluent, has reached epidemic proportions. Health care in the public sector in these countries is organized in the shape of a pyramid, with primary health centers at the bottom, followed by intermediate-level hospitals and referral hospitals at the top. Specialized care for disease is usually available only at the major referral hospitals. There are only about 900 nephrologists in India. There is a thriving private sector health care industry, but treatment costs are quite high, and only the rich or patients whose expenses are covered by their employers can afford treatment in these hospitals.

The differences in living standards and availability of health care determine the variations in disease patterns, management practices, and disease outcome. The spectrum of kidney

Table 79.1 Economic and Development Indicators on the Indian Subcontinent

Parameter	India	Pakistan	Bangladesh	Sri Lanka
Population (billions)	1.22	0.193	0.163	0.021
Population growth rate (%)	1.31	1.55	1.58	0.91
Life expectancy at birth (yr)	67.48	66.7	70.36	76.15
Median age (yr)	26.7	22.2	23.9	31.4
Percentage of total population that is elderly (>65 yr)	5	4	5	8
Literacy rate (%)	64.84 (in 2011)	55 (in 2009)	57.7 (in 2011)	92.5 (in 2010)
Infant mortality rate/1000 live births (in 2013)	44.6	59.35	47.30	9.24
Per capita gross domestic product purchasing power parity (U.S. \$; in 2012)	3800	3100	2000	6000
Percentage of the population living below the national poverty line	29.8	22.3	31.5	8.9
Human development index	0.554	0.515	0.515	0.715
Country ranking by human development index (of 179 countries)	136	146	146	92

Table 79.2 Causes of Acute Kidney Injury in a Tertiary Referral Center on the Indian Subcontinent (%)

Cause	1965–1974 ^a	1975–1980 ^a	1981–1986 ^a	2001–2006 ^b
Medical causes	67	55	61	65
Diarrheal diseases	23	12	10	7.3
Intravascular hemolysis due to glucose-6-phosphate dehydrogenase deficiency	12	12	6	1
Glomerulonephritis	11	9	9.5	11
Copper sulfate poisoning	12	12	6	1
Chemical and drug exposure	4	5	7	7.2
Snakebites and insect stings	3	3	2.5	6.5
Sepsis	0	3	4	17.8
Miscellaneous	7	11	17	13.2
Obstetric causes	22	21	9	5
Surgical causes	11	24	30	30

^aAdapted from Chugh KS, Sakhuja V, Malhotra HS, et al. Changing trends in acute renal failure in third world countries—Chandigarh study. *Q J Med.* 1989;73:1117–1123.

^bAuthors' unpublished data.

diseases in this region is characterized by a mix of conditions that are globally encountered and those specific to the subcontinent. The latter can be secondary to a genetic predisposition in specific ethnic groups or related to exposure to environmental factors, such as climatic conditions, infectious agents, envenomation, and chemical toxins. Genetic factors that predispose to kidney disease in the tropics include glucose-6-phosphate dehydrogenase (G6PD) deficiency, which gives rise to intravascular hemolysis and pigment-induced acute kidney injury (AKI). Indigenous health care systems are still popular in rural areas, and patients are frequently treated with herbs and potions that could add to the burden of kidney disease.

ACUTE KIDNEY INJURY

Reliable statistics on the patterns and prevalence of AKI in this region are not available, but it is the most commonly encountered renal emergency.^{4,6} For example, about 0.1% to 0.25% of all admissions to the authors' hospital admissions are for AKI. In a study of 536 patients admitted in a pediatric

intensive care unit (ICU) in a large tertiary care hospital in North India, AKI was seen in 42.9% of subjects.⁷ However, it was lower, at 25.1%, in South India under similar clinical settings.⁸ Most of the causes of AKI described elsewhere in the world are encountered in the Indian subcontinent as well. Several causes, however, either are unique or are seen with increased frequency in this part of the world. Compared with industrialized countries, in which AKI is mainly a disease of the elderly and is seen primarily in hospitalized patients, community-acquired AKI in otherwise healthy individuals is common in countries in this region, a characteristic they share with other developing countries.⁹ Patients with AKI are younger in this region than their counterparts in the West. The median age of patients with AKI in the West increased from 41.2 years in the 1950s to 60.5 years in the 1980s¹⁰; the average age of patients with AKI in India is 37.1 years.⁹ Hospital-acquired AKI is more common in the older population than in the young.¹¹

Over the years, there has been a change in the spectrum of AKI (Table 79.2).¹² In urban areas, the causes of AKI are similar to those in the industrialized world, but in rural areas, AKI is mainly the result of diarrheal illnesses, chemicals,

snake bite, and insect stings.¹³ Renal failure after the suicidal ingestion of copper sulfate has shown a significant decline,^{14,15} possibly because of the easy availability of other poisons, such as organophosphate insecticides.¹⁶ At a large tertiary care center in Pakistan, paraphenylene diamine poisoning was the most common cause for poisoning associated with AKI.¹⁷ Data from a large tertiary care referral institute in India has shown that sepsis as a cause of AKI has increased almost seven times, from just 1.57% from 1983 through 1995 to 11.43% from 1996 to 2008.¹⁸

Treatment facilities are grossly inadequate. Patients are referred late, only after renal failure becomes severe and complications set in. They often require immediate dialysis and, therefore, the mortality rate is high. Intermittent peritoneal dialysis (PD) is still used in several areas¹⁹⁻²¹ because hemodialysis (HD) facilities are limited to bigger cities. The outcome of AKI is worse in the elderly than in the younger population.²² Conditions causing AKI that are specific to the Indian subcontinent are discussed in the following sections.

PREGNANCY-ASSOCIATED ACUTE KIDNEY INJURY

Improvements in obstetric care have led to the virtual disappearance of pregnancy-related AKI in the industrialized world. Data from a large tertiary care hospital in the eastern part of India have shown that the proportion of obstetric AKI cases has remained more or less similar at 11.8% and 12.8% of total cases of AKI over two 13-year time periods (1996–2008 vs. 1983–1995, respectively).²³ However, some investigators have described a decline in the rate of obstetric AKI in India from 22% to 5% of all cases of AKI over the past 4 decades (Table 79.2).¹⁹ The frequency distribution of AKI is bimodal in terms of the duration of gestation.

The first peak, seen between 8 and 16 weeks of gestation, is associated chiefly with induced septic abortions. The abortion practices include the use of sticks, the insertion of abortifacient chemicals, pastes, and soap solutions, and dilation and curettage performed under unhygienic conditions and by untrained personnel.²⁴ Postabortal sepsis has remained the most common cause for obstetric AKI over the past 3 decades.²³ The incidence of AKI has decreased since the legalization and regulation of abortions and the wider availability of medical facilities,^{24,25} although heterogenous, limited data from other countries have also suggested that postabortal or puerperal sepsis is the leading cause of obstetric AKI.²⁶

The second peak in obstetric AKI occurs after 34 weeks of gestation and is related to preeclampsia, eclampsia, abruptio placentae, postpartum hemorrhage, and puerperal sepsis. The incidence of preeclampsia in the Indian population has been reported to be 28.7% in a population-based study.²⁷ Data from Pakistan, Bangladesh, and Nepal have also suggested sepsis related to abortion or puerperium, hemorrhage, hypotension, and preeclampsia as main associations of pregnancy-related AKI.²⁶ Acute cortical necrosis has been observed in about 25% of patients with obstetric AKI in parts of India.^{23,24,27} The provision of institutional care and early referral for patients with complicated pregnancies to specialized centers is important to ensure better prevention and management of AKI in such settings. Community-based education and awareness are required, so that patients seek proper medical attention as soon as possible. AKI in pregnancy is managed along the

usual lines, with additional cause-specific interventions, if any.²⁶ Those women who do not recover kidney function usually undergo contrast-enhanced computed tomography (CT) of the kidneys to look for acute cortical necrosis. If this is absent, a renal biopsy is done in such individuals. Pregnancy and kidney disease are further discussed in Chapter 48.

ACUTE KIDNEY INJURY ASSOCIATED WITH SURGICAL CAUSES

The proportion of AKI from surgical causes has increased from 11% to 30% (see Table 79.2). Obstructive nephropathy constitutes a major cause of surgical AKI.²⁸ The high incidence of nephrolithiasis contributes and is related to inherited metabolic disorders, dietary factors, and fluid losses. A high prevalence of nephrolithiasis in some areas, aggravated by dehydration caused by a combination of exposure to high ambient temperatures and a high prevalence of diarrheal diseases, is an example. Faith in the efficacy of indigenous medicines to dissolve stones causes delay in surgical intervention and hastens the development of renal failure.

ACUTE KIDNEY INJURY DUE TO INFECTIONS

DIARRHEAL DISEASES

AKI secondary to diarrhea is encountered not only in children but also in adults. In adults, diarrheal diseases cause 30% of all AKI,⁹ whereas 35% to 50% of all children undergoing dialysis for AKI have preceding diarrhea and dehydration.²⁸⁻³⁰ This problem is common in rural areas and urban slums inhabited by poverty-stricken individuals, where sanitation is poor and potable water is not available. The incidence increases during summer and the rainy season. Associated vomiting is an early feature of rotavirus infection. Loose watery stools indicate infection with enterotoxigenic *Escherichia coli* or *Vibrio cholerae*. Fever, cramps, tenesmus, and blood and mucus in stools suggest *Shigella*, *Salmonella*, or enteroinvasive *E. coli* infection. The diagnosis of cholera can be confirmed by microscopic demonstration of the highly motile *V. cholerae* organisms in a hanging drop preparation; culture is necessary to confirm the presence of other organisms.

Early and adequate fluid replacement is the cornerstone of therapy. Widespread use of the oral rehydration solution recommended by the World Health Organization (WHO) has led to a significant decline in the mortality rate from this condition. Intravenous rehydration with lactated Ringer solution may be required in patients with severe dehydration, persistent vomiting, or paralytic ileus. Hypokalemia may worsen as the metabolic acidosis is corrected, and large amounts of potassium may be required to prevent life-threatening cardiac arrhythmias. Because peritoneal dialysis fluid is potassium-free, potassium should be replaced through an intravenous (IV) or intraperitoneal route in patients with severe AKI treated with this modality. Volume depletion secondary to acute gastroenteritis is presumed to be the most common cause of kidney injury; occasionally, acute cortical necrosis can also occur.³¹

HEMOLYTIC-UREMIC SYNDROME

Hemolytic-uremic syndrome was responsible for 25% to 55% of all cases of pediatric AKI in some parts of the subcontinent, but the proportion has decreased.^{8,32-34} In

one report, diarrhea-related, hemolytic-uremic syndrome accounted for only 3.8% of pediatric AKI cases at a pediatric ICU in South India.⁸ This condition is seen mainly in preschool-aged children and is less common in adults. The main feature is oliguric renal failure, preceded by a diarrheal prodrome in about 70% of cases. Neurologic involvement is seen in 30% to 50% of cases. Examination shows pallor and mild icterus. Renal failure is severe and requires prolonged dialysis. Diagnosis is confirmed by the demonstration of fragmented erythrocytes on blood smear, as well as thrombocytopenia. Supportive evidence includes unconjugated hyperbilirubinemia and increased plasma lactate dehydrogenase levels. The causative organism in the Indian subcontinent is usually *Shigella* and not *E. coli*.^{35,36} The histologic hallmark of this condition is thrombotic microangiopathy in the renal vasculature. Histologic examination shows frequent involvement of arterioles and small arteries, with severe intimal proliferation and luminal stenosis. Patchy or diffuse renal cortical necrosis develops in up to 40% of patients.³² The treatment is mainly supportive. Plasma infusions or plasma exchange are used infrequently; the outcome is poor, with a mortality rate of 60%.³⁷ Of those who recover, a significant proportion are left with residual renal dysfunction and eventually progress to ESKD. The management of thrombotic microangiopathies is discussed in detail in Chapter 34.

MALARIA

Malaria is caused by the protozoan *Plasmodium* and is transmitted by the *Anopheles* mosquito. Of the four *Plasmodium* species, AKI is seen most frequently with *Plasmodium falciparum* infection. A few reports have described an association of AKI with *Plasmodium vivax* infection.^{38,39} Multiple organ dysfunctions have been also associated with *P. vivax* malaria.⁴⁰

These forms of malaria predominate in warmer regions closer to the equator in the subcontinent (parts of India, Pakistan, Bangladesh, and Sri Lanka) and are associated with intense, year-round transmission. In addition to being a burden to the native communities, malaria is a danger to nonimmune travelers to endemic areas.⁴¹ The Southeast Asian region has the highest burden of malaria after Africa. India, Indonesia, Pakistan, and Myanmar are the leading countries that are affected.⁴¹ A more than 50% reduction in incidence is expected over the next few years as compared with the incidence in 2000.⁴² In 2016, Sri Lanka was declared malaria-free by WHO.⁴³ Preventive measures are directed at reducing exposure and preventing transmission of disease by better vector control and sanitation. Travelers from nonendemic areas should be aware of the risks and encouraged to seek treatment or chemoprophylaxis whenever indicated.

The contribution of malaria to overall hospital admissions for AKI varies from 2% to 39%.⁴⁴ The incidence of malarial AKI among those living in endemic areas is 2% to 5%, but 25% to 30% of nonimmune visitors with malaria experience renal failure.⁴⁵ Among those with severe parasitemia, as many as 60% have AKI.⁴⁶ In Pakistan, malaria-induced AKI also contributes significantly to the total AKI burden.⁴⁷

Clinical Features

Malaria causes classic paroxysms of spiking fever, along with malaise, myalgia, headache, and chills, and the disease can be confused with a viral illness. Nausea, vomiting, and

hypotension are common in nonimmune individuals. Severe infection may involve several vital organs, including the central nervous system, manifesting as deep coma and seizures, noncardiogenic pulmonary edema, shock, and disseminated intravascular coagulation (DIC). AKI usually is seen by the end of the first week and is nonoliguric in 50% to 75% of cases.⁴⁵ In more than 75% of cases, AKI is associated with cholestatic jaundice. It is usually associated with a hypercatabolic state, with a rapid increase in creatinine levels. In a study from India, of a total of 101 patients with malarial AKI over a 1-year period, about 37% required dialytic support, and hospital mortality was around 10%.⁴⁸ Other manifestations include nonnephrotic proteinuria, microscopic hematuria, hemoglobinuria, and electrolyte abnormalities.

Diagnosis can be established by the demonstration of asexual forms of the parasite in peripheral blood smears stained with Giemsa stain. Staining with the fluorescent dye acridine orange allows more rapid diagnosis. Also, simple but specific antibody-based card tests that detect *P. falciparum*-specific histidine-rich protein 2 or lactate dehydrogenase antigens in fingerstick blood samples have been introduced. Most patients present with fever and nonspecific symptoms and are often treated empirically, depending on which are the most common local causes for acute febrile illness. These simple card-based tests allow rapid bedside diagnosis and early institution of therapy. They allow differentiation of *P. falciparum* from non-*P. falciparum* malarias, but results can remain positive for several weeks after acute infection. A scoring system based on the number of organ systems involved has been suggested for judging the severity and predicting outcomes in *P. falciparum* malaria.⁴⁹

Histologic Features

Acute tubular necrosis is the most common lesion found on histologic analysis. Pigment casts may be seen in tubular lumina in patients with intravascular hemolysis. Varying degrees of interstitial edema and mononuclear cell infiltrate are common accompaniments.⁴⁹

Pathogenesis

Disease severity is related to the intensity of infection.⁵⁰ The most important pathogenetic factor is the hemodynamic change caused by altered erythrocyte rheology. The infected cell becomes more spherical and less deformable as a result of the formation of membrane protuberances or knobs on the cell surface. These knobs extrude a strain-specific adhesive variant protein of high molecular weight that mediates attachment to receptors on venular and capillary endothelium, causing a phenomenon called “cytoadherence.” The major adhesive protein family is known as “*P. falciparum* erythrocyte membrane protein.” The cell surface receptors for these knobs include complement receptor 1, glycosaminoglycans, intercellular adhesion molecule-1, chondroitin sulfate B, CD36, platelet endothelial cell adhesion molecule 1/CD31, thrombomodulin, E- and P-selectins, and vascular cell adhesion molecule-1. Some of these receptors are expressed constitutively, whereas others are induced by inflammatory mediators released in severe disease. Infected erythrocytes also adhere to uninfected red cells, platelets, monocytes, and lymphocytes. These aggregated and sequestered red blood cells interfere with microcirculatory flow in the kidneys and other organs.⁵¹ This phenomenon is unique to *P. falciparum*

infection and has not been observed in infection with other species.

Treatment

Mortality from malarial AKI varies between 15% and 50%.⁴⁶ Severe *P. falciparum* malaria requires intensive nursing and close multidisciplinary care.⁴⁶ Prompt assessment of volume status, blood glucose level, and acid–base status is essential. All patients with *P. falciparum* infection should be presumed to have chloroquine-resistant infection. The cinchona alkaloids (quinine or quinidine) are the mainstay of treatment because of their activity against chloroquine-resistant strains. Quinine often causes hyperinsulinemia and hypoglycemia, and many centers recommend the administration of a continuous infusion of 5% to 10% dextrose to all patients.⁵² Compounds derived from artemisinin (isolated from the Chinese herb *Artemisia annua*) such as artesunate, artemether, and arteether are particularly valuable in areas with quinine resistance and in patients with recurrent quinine-induced hypoglycemia.

All patients should also receive gametocidal therapy (tetracycline or pyrimethamine–sulfadoxine). Mefloquine, halofantrine, atovaquone, artemisinin, and pyrimethamine–sulfadoxine (Fansidar) derivatives are possible alternatives for resistant *P. falciparum* malaria. Patients with evidence of hemolysis should receive adequate hydration and parenteral sodium bicarbonate to alkalinize the urine to a pH of more than 7.0. AKI is usually associated with a hypercatabolic state, and frequent dialysis may be needed.⁵³ Hyperkalemia should be watched for and adequately treated. The peritoneal microcirculation is impaired as a result of clogging with infected erythrocytes and vasoconstriction, reducing the efficacy of PD. Despite these limitations, PD has been shown to improve survival by 16%, with early institution of manual, immediate PD in AKI due to *P. falciparum* malaria.⁵⁴ Exchange transfusion has also been tried in patients with severe parasitemia.

LEPTOSPIROSIS

Leptospirosis, the most widespread zoonosis in the world,⁵⁵ is particularly prevalent in parts of southern and western India.^{56,57} The animal hosts include rats, mice, gerbils, hedgehogs, foxes, dogs, cattle, sheep, pigs, and rabbits. Even asymptomatic animals carry a high number of organisms ($>10^{10}$ /g) in their kidneys and shed *Leptospira* in urine for years.⁵⁸ Human infection occurs incidentally, either directly through contact with the urine or tissue of an infected animal or indirectly through contaminated water, soil, or vegetation. The usual portals of entry are abraded skin and exposed mucosae. Leptospirosis is an occupational hazard for coal miners and sewage, abattoir, and farm workers, as well as workers in the aquaculture industry. The genus *Leptospira* includes the pathogenic *Leptospira interrogans* strains and the saprophytic *Leptospira biflexa*. The disease occurs throughout the year, with an increase in incidence during or soon after the rainy season, especially after floods. Adult men are affected most frequently.

Clinical Features

The manifestations of leptospirosis vary from subclinical infection or a self-limited anicteric febrile illness to severe and potentially fatal disease.^{55,56} Symptoms appear 1 to 2 weeks after exposure and are typically biphasic in character. The leptospiremic phase is characterized by high fever, with chills,

headache, and severe muscle aches and tenderness. Renal failure develops in the second, immune phase, when patients also have progressive jaundice, epistaxis, hemoptysis, gastrointestinal bleeding, hemorrhagic pneumonia, and bleeding into the adrenal glands. The triad of acute renal failure, cholestatic jaundice, and bleeding constitutes Weil syndrome caused by Leptospirosis. AKI occurs in 20% to 85% of cases.^{57,59,60} In about 50% of cases of AKI, renal failure is associated with polyuria and hypokalemia, along with increased fractional excretion of potassium. Renal failure is mild and nonoliguric in patients without icterus. Hypotension is noted in more than 60% of cases and is often unresponsive to volume expansion and inotropic support. Hypotensive patients are also more likely to have adult respiratory distress syndrome.⁶¹

Diagnosis is based on culture or serologic testing. Organisms can be isolated in blood cultures during the first phase and later from urine. However, growth takes up to 4 weeks in Fletcher or Stuart semisolid medium. The macroscopic agglutination test can be used as a screening test but is not specific. The benchmark is the microscopic agglutination test, but this is a complex procedure that requires maintenance of live *Leptospira* cultures. An immunoglobulin M (IgM)–specific dot enzyme-linked immunosorbent assay has been found to be specific in diagnosing leptospirosis in endemic areas. Urinalysis during the leptospiremic phase reveals mild proteinuria, as well as hyaline and granular casts.

Histologic Features

The kidneys typically are swollen and may be bile-stained. The main lesions seen on light microscopy are interstitial edema and infiltration with mononuclear cells and eosinophils. Mild and transient mesangioproliferative glomerulonephritis with C3 and IgM deposition may sometimes be observed.⁶²

Pathogenesis

Renal involvement results from direct invasion of the renal tissue by the organism, which leads to liberation of enzymes, metabolites, and endotoxins, as well as complement activation. Ultrastructural studies after inoculation of *Leptospira pomona* into mice have shown the organism infiltrating the glomerular capillary lumen at day 2, the interstitium at days 4 to 8, the proximal tubular cells by day 10, and the tubular lumen by day 14.⁶³ Several leptospiral outer membrane proteins have been localized to the proximal tubules and interstitium of infected animals. The addition of *Leptospira* endotoxin to human macrophages induces tumor necrosis factor- α (TNF- α). The glycoprotein component of endotoxin could inhibit renal Na⁺-K⁺-adenosine triphosphatase (Na⁺-K⁺-ATPase), in turn affecting the apical Na-K-2Cl cotransporter and leading to potassium wasting.⁶⁴ An upregulation of nuclear factor kappa B binding to DNA was noted on the addition of outer membrane extracts from pathogenic serovars to cultured medullary thick ascending limb cells; this was accompanied by an increase in inducible nitric oxide synthase, monocyte chemoattractant protein-1, and TNF- α . Alterations in intravascular volume, hemoglobinuria, and myoglobinuria also contribute.⁶² The tubules become insensitive to the action of antidiuretic hormone.

Treatment

Leptospirosis is a self-limiting disease, and patients with mild cases recover spontaneously. The emphasis is on symptomatic

measures, together with correction of hypotension and fluid and electrolyte imbalances. Administration of crystalline penicillin or doxycycline shortens the duration of fever and hospital stay and may hasten amelioration of leptospirosis. In a meta-analysis of controlled trials, the role of various antibiotics for the treatment of leptospirosis was found to be uncertain; this finding could be attributed to a lack of adequate clinical trials.⁶⁵ The role of penicillin in the treatment of leptospirosis can be debated.⁶⁵ Patients with renal failure need close monitoring and dialysis when necessary. Poor prognostic indicators include increasing age and the presence of jaundice, pulmonary complications, hyperbilirubinemia, diarrhea, hyperkalemia, pulmonary rales, and hypotension on admission.⁶⁶ Internal hemorrhage and myocarditis are common causes of death.

ZYGOMYCOSIS

Zygomycosis is a rare opportunistic infection caused by fungi of the order Mucorales and genera *Rhizopus*, *Absidia*, and *Rhizomucor*. Organ involvement occurs through vascular invasion, which leads to thrombosis of large and small arteries and infarction and necrosis of the affected organ. The major presentations have rhinocerebral, pulmonary, gastrointestinal, and disseminated forms. Renal mucormycosis involving the major renal vessels has been reported primarily in northern India in the past decade and may occur as isolated renal involvement or as part of disseminated disease.⁶⁷ The administration of IV fluids contaminated with fungus may be responsible in some patients. Bilateral renal arterial involvement leads to oliguric AKI. The condition usually develops in otherwise immunocompetent individuals; presentation is with fever, lumbar pain, pyuria, and oliguria. CT reveals enlarged nonenhancing kidneys with perirenal collections and/or intrarenal abscesses (Fig. 79.1).⁶⁸ The diagnosis can be confirmed by demonstration of hyphae in the material obtained by aspiration or percutaneous biopsy. The only definitive treatment is extensive débridement of affected tissue, which may include bilateral nephrectomy, and



Fig. 79.1 Renal mucormycosis. This contrast-enhanced computed tomography scan shows bulky kidneys, with large nonenhancing areas (arrows) involving the cortex and medulla and with no contrast in the pelvicalyceal system.

systemic antifungal therapy with amphotericin B. Bilateral renal mucormycosis carries an extremely poor prognosis.⁶⁷

DENGUE FEVER

Dengue fever is the most prevalent mosquito-borne viral disease. There are four closely related but serologically distinct dengue viruses—DENV-1, DENV-2, DENV-3, and DENV-4, of the genus *Flavivirus*. The infection is primarily transmitted by *Aedes* mosquitoes, particularly *Aedes aegypti*. Dengue fever is a self-limiting illness, with symptoms including fever, headache, and muscle and joint pain. However, a life-threatening dengue hemorrhagic fever (DHF) develops in a small minority of patients, with low platelet count, bleeding, and capillary plasma leakage.⁶⁹ In severe cases, patients may have dengue shock syndrome. The cardinal features of DHF are increased vascular permeability (plasma leakage syndrome), marked thrombocytopenia, hemorrhagic tendency (as demonstrated by a positive tourniquet test result), and spontaneous bleeding.

In a study from South India, AKI was seen in 10.8% of patients with dengue fever. Of these, 12 (5.4%) had mild AKI, 7 (3.1%) had moderate AKI, and 5 (2.2%) had severe AKI according to Acute Kidney Injury Network (AKIN) criteria.⁷⁰ Proposed mechanisms for AKI are capillary leak, shock resulting in hypoperfusion, and acute tubular necrosis. Also, hemolysis or rhabdomyolysis contribute to AKI. Of 198 patients with dengue admitted to an ICU, 15 (7.6%) required dialytic support.⁷¹ The presence of dengue hemorrhagic and dengue shock syndrome, neurologic involvement, and prolonged activated partial thromboplastin time have been found to be independent predictors of AKI. Mortality is about 6% to 9% in patients with dengue fever and AKI. AKI during hospitalization for dengue fever is a predictor of death.⁷² Vector control, improvements in sanitation, and preventing exposure are the main strategies for prevention of this disease.

ACUTE KIDNEY INJURY DUE TO SNAKE BITE AND INSECT STINGS

SNAKE BITE

Snake bite is an occupational hazard and occurs when people are working barefoot in the fields.^{73,74} On the Indian subcontinent, AKI develops after bites by snakes of the viper family, such as Russell's viper, the saw-scaled viper (*Echis carinatus*), and pit vipers. The incidence of AKI varies from 13% to 32% in India after Russell's viper or *E. carinatus* bites⁷⁵⁻⁷⁷ and is 27% in Sri Lanka after bites of unidentified vipers.⁷⁸

Clinical Features

The severity of the symptoms and signs is related to the type of venom, as well as to the dose injected during the bite. Severely envenomed patients experience DIC, which frequently results in spontaneous bleeding and coagulopathy. The latter features often dominate the clinical course until coagulation returns to normal.⁷⁹ Many viper-envenomed patients have hypotensive shock as a consequence of hypovolemia from significant blood loss. Blood can ooze continuously from the fang marks, and severe hemorrhage may manifest as hematemesis, melena, hemoptysis, or bleeding into the muscles, fascial compartments, serous cavities, and subarachnoid space.⁸⁰ Pain and swelling of the bitten part are generally the earliest symptoms, appearing within a few minutes. The swelling may spread to involve the whole limb

and is due to exudation of plasma or extravasation of blood into the subcutaneous tissues. Blistering or local necrosis is observed in one-third to one-half of patients.

In patients with AKI, oliguria often develops rapidly, within the first 24 hours, but it may be delayed until 2 to 3 days after the bite. Some patients become anuric, whereas occasional patients remain nonoliguric. Urine may show gross or microscopic hematuria. Some patients complain of pain in the renal angle preceding oliguria, which may be a useful clue to impending renal failure. Jaundice and hemoglobinuria resulting from intravascular hemolysis are not infrequent after Russell's viper or *E. carinatus* bites and have been reported in India and Sri Lanka.⁷⁷

Laboratory investigations reveal varying degrees of anemia, resulting from a combination of intravascular hemolysis and blood loss. Hemolysis results in unconjugated hyperbilirubinemia, reticulocytosis, elevated plasma free hemoglobin level, and hemoglobinuria. In some patients, the peripheral blood smear may show fragmented erythrocytes, suggesting microangiopathic hemolysis. The blood fails to coagulate normally, and features of DIC are often present.⁸⁰ Thrombocytopenia may occur, however, even in the absence of a consumptive coagulopathy.

The mortality rate for patients with snake bite–associated AKI varies with the nature of the renal lesion. According to one report, although only 16% of those with acute tubular necrosis died, as many as 80% of those with cortical necrosis had a fatal outcome.⁷⁷

Histologic Features

Renal histologic examination shows acute tubular or cortical necrosis. Glomerular changes have been described in rare cases, but their significance is not known.

Acute Tubular Necrosis. Acute tubular necrosis is the predominant lesion seen in 70% to 80% of patients with snake bite–associated AKI. On light microscopy, the tubules appear dilated and lined by flattened epithelium. Severe cases exhibit cell necrosis and desquamation of necrotic cells from the basement membrane. Hyaline, granular, or pigment casts are seen in tubular lumina. Varying degrees of interstitial edema, hemorrhage, and inflammatory cell infiltration are present. Later biopsy specimens reveal regenerating tubular epithelium. Intrarenal blood vessels are usually unaffected.

On ultrastructural examination, proximal tubules show dense intracytoplasmic bodies that represent degenerating organelles or protein resorption droplets.⁸¹ Small areas of basement membrane are denuded. Distal tubular cells have a dilated endoplasmic reticulum and many degenerating organelles. Apoptosis, a prominent feature in the distal tubules, indicates a high cell turnover. In the interstitium, fibroblasts appear active, with increased numbers of organelles and cytoplasmic processes. Mast cells and eosinophils show both granulated and partially degranulated forms.

Acute Cortical Necrosis. Bilateral diffuse or patchy cortical necrosis has been observed in 20% of patients with snake bite–induced AKI.⁷⁷ The presence of fibrin thrombi in the arterioles is a prominent feature in these patients. A narrow subcapsular rim of cortex often escapes necrosis. The area underlying this rim, however, shows necrosis of

glomerular as well as tubular elements. The necrotic zone is often bordered by an area of hyperemia and leukocytic infiltration.

Pathogenesis

The exact pathogenesis of AKI following snake bite is not well established because of the lack of a reproducible animal model. However, a number of factors may contribute, including bleeding, hypotension, circulatory collapse, intravascular hemolysis, DIC, microangiopathic hemolytic anemia, and the direct nephrotoxicity of the venom. Bleeding either into tissues or externally and loss of plasma into the bitten extremity can produce hypotension and circulatory collapse. These effects are caused by venom metalloproteinases that degrade basement membrane proteins surrounding the vessel wall, leading to loss of integrity. Hemorrhagic toxins have been isolated from the venom of many snakes of the Viperidae and Crotalidae families.⁷⁹ In addition, vasodilation and increased capillary permeability can aggravate the circulatory disturbances of shock.

Hemolysis results from the action of phospholipase A₂, which is present in almost all snake venoms. Phospholipase A₂ causes hemolysis directly by hydrolysis of red blood cell membrane phospholipids or indirectly via the production of the strongly hemolytic lysolecithin from plasma lecithin.⁸² The human hemostatic system is regulated through a number of critical interactions involving blood proteins, platelets, endothelial cells, and subendothelial structures. Snake venoms, particularly those from the viper and pit viper families, contain many proteins that interact with members of the coagulation cascade and the fibrinolytic pathway.⁸²

Treatment

The therapeutic approach to renal failure after snake bite is the same as that for AKI from any other cause. Early administration of antivenom is vital; delay results in a steep increase in the dose requirements. Indications include prolonged coagulation time or failure of blood coagulation, spontaneous systemic bleeding, intravascular hemolysis, and local swelling involving more than two segments of the bitten limb. Knowledge of the offending snake species allows administration of monovalent antivenom, if available. Immunodiagnostic techniques are helpful in the identification of the venom antigen. The enzyme-linked immunosorbent assay has been used extensively in rural Thailand for this purpose.⁸³ The currently available test, however, is not quick enough for clinical application. Because only polyvalent antivenom is available in most parts of Asia, precise identification of the snake is not essential for management.

Indian studies recommend initial administration of 20 to 100 mL of antivenom, followed by repeated doses of 25 to 50 mL every 4 to 6 hours until the effects of systemic envenomation disappear.^{80,84} A simple way to monitor antivenom treatment efficacy is by monitoring whole blood clotting time three or four times daily. Coagulation is generally restored within 6 hours of an adequate dose. Coagulation testing must be performed for at least 3 more days, because delayed absorption of the venom can lead to recurrence of the coagulopathy. Immunoassays permit serial estimation of venom levels and are useful in guiding antivenom therapy.

Other therapeutic measures include replacement of lost blood with fresh blood or plasma, maintenance of electrolyte balance, administration of tetanus immunoglobulin, and treatment of pyogenic infection with antibiotics. The prognosis is good in patients who receive adequate doses of antivenom. One study has shown the outcome for venom-induced AKI to be better after HD than after PD.⁸⁵ The overall mortality rate is about 30%.⁷⁷

BEE, WASP, AND HORNET STINGS

Honeybees, yellow jackets, hornets, and paper wasps are stinging insects belonging to the order Hymenoptera. An isolated sting causes just a local allergic reaction, but attack by a swarm of insects introduces a large dose of the venom, sufficient to cause systemic symptoms.⁸⁶⁻⁸⁸ These include vomiting, diarrhea, hypotension, loss of consciousness, and AKI. Patients with AKI have been reported to have received from 22 to more than 1000 stings.⁸⁹ AKI is secondary to hemolysis and/or rhabdomyolysis.⁸⁹ Hemolysis results from the action of a basic protein fraction, as well as melittin and phospholipase A present in the venom. Rhabdomyolysis has been attributed to polypeptides, histamine, serotonin, and acetylcholine. A direct nephrotoxic role for these venoms has also been suggested. Renal biopsy specimens reveal acute tubular necrosis.

ACUTE KIDNEY INJURY DUE TO CHEMICAL TOXINS

COPPER SULFATE POISONING

Copper sulfate is a strong corrosive that produces symptoms within minutes of ingestion, usually with suicidal intent. Metallic taste, excessive salivation, burning retrosternal and epigastric pain, nausea, and repeated vomiting are the initial features. The vomitus is blue-green and turns deep blue on the addition of ammonium hydroxide, which allows it to be differentiated from bile. Diarrhea, hematemesis, and melena follow. Jaundice, hypotension, convulsions, and coma may develop in severe cases. Acute pancreatitis, myoglobinuria, and methemoglobinemia have also been reported. AKI is seen in 20% to 25% of cases and is invariably oliguric.⁹⁰ Hemoglobinuria may be seen in about 40% of cases. Diuresis ensues after 7 to 10 days and is usually followed by complete renal recovery.¹⁴

Copper can produce considerable oxidative stress and interferes with the activity of several key enzymes, such as $\text{Na}^+\text{-K}^+\text{-ATPase}$, G6PD, glutathione reductase, and catalase.⁹⁰ Direct nephrotoxicity, severe hemolysis, and hypovolemia secondary to fluid loss are the main factors responsible for kidney injury. In experimental animals, copper sulfate causes toxic damage to the proximal tubules.⁹⁰ Histologic examination usually shows acute tubular necrosis, with predominant involvement of the proximal tubules. Hemoglobin casts may be seen in patients with intravascular hemolysis.^{90,91} Acute cortical necrosis has been seen rarely.

Management entails gastric lavage using 1% potassium ferrocyanide, which leads to the formation of insoluble cupric ferrocyanide. Egg whites or milk can be administered as an antidote. Emesis should not be induced. Any volume deficit should be corrected quickly. Hyperkalemia may be severe and sustained because of the ongoing hemolysis and requires early and frequent dialysis.

ETHYLENE GLYCOL POISONING

Diethylene and polyethylene glycols have been used as cheap substitutes for propylene glycol as vehicles in pediatric syrup preparations. Epidemics of diethylene glycol-induced AKI have been reported in India and Bangladesh.⁹²⁻⁹⁴ In one large study, 236 deaths were recorded among 339 children with unexplained AKI in a children's hospital in Dhaka, Bangladesh.⁹² A total of 51 children had ingested a brand of acetaminophen (paracetamol) known to contain diethylene glycol, whereas 85% of the remaining patients had ingested an unknown elixir for fever.⁹² The likelihood of hepatomegaly, edema, hypertension, severe acidosis, higher serum creatinine level, and in-hospital mortality was higher in patients with suspected poisoning.⁹² In another report, 14 patients died of AKI after the administration of glycerol to decrease intracranial or intraocular pressure.⁹³ Analysis of this preparation showed it to be 70% ethylene glycol. Autopsy revealed acute cortical necrosis as the most frequent lesion.

ETHYLENE DIBROMIDE POISONING

Ethylene dibromide (EDB), a pesticide fumigant, is absorbed from the skin, gastrointestinal tract, and intestinal mucosa. Both accidental and suicidal poisonings with EDB have been reported. AKI and hepatocellular injury are the chief manifestations.^{95,96} The mortality remains very high, despite all supportive measures.⁹⁷ EDB is postulated to lead to the generation of free oxygen radicals through the cytochrome P450 pathway, causing lipid peroxidation and membrane damage. Dimercaprol has been suggested as an antidote on the basis of the structural similarities of the two compounds.

CHROMIC ACID POISONING

Chromic acid (H_2CrO_7) and its salts (chromates and dichromates) are used in the electroplating, leather tanning, and anticorrosive metal treatment industries. Renal lesions have been reported after the acute ingestion of large quantities of these substances.^{98,99} Ingestion is most often with suicidal intent and is followed by severe abdominal pain, vomiting, gastrointestinal bleeding, and circulatory collapse. Dichromate is directly nephrotoxic and causes extensive proximal tubular necrosis.¹⁰⁰ Hypotension and hemolysis also contribute to tubular damage. Management entails gastric lavage with alkaline solutions such as sodium bicarbonate to prevent absorption and IV fluids to combat hypotension. Forced diuresis enhances renal excretion of the compound. Reducing agents such as vitamin C have been shown to prevent chromic acid-induced acute tubular necrosis in experimental animals.¹⁰¹

HAIR DYE-RELATED ACUTE KIDNEY INJURY

Hair dye and its constituent, paraphenylenediamine (PPD), have been reported as an accidental and intentional cause of poisoning on the Indian subcontinent.^{102,103} Acute poisoning with PPD due to accidental or intentional consumption causes edema of the face and neck, resulting in severe respiratory distress that requires tracheostomy. The preceding manifestations are followed by rhabdomyolysis and AKI. In one study, AKI was seen in 32% of patients consuming hair dye.^{102,103} The mean nephrotoxic dose was 79 mL.¹⁰² Mechanisms of AKI

are direct nephrotoxicity, myoglobinuria, hemoglobinuria, hypovolemia, and hypotension.^{102,103} The renal lesions are usually found to be acute tubular necrosis or acute interstitial nephritis in patients undergoing biopsy. Renal replacement therapy (RRT) is required in 82% of patients in whom AKI develops.¹⁰² The overall mortality rate is about 26%, and death is attributed mainly to acute respiratory distress, cardiac arrhythmias, and shock.¹⁰³

ACUTE KIDNEY INJURY DUE TO INTRAVASCULAR HEMOLYSIS AND GLUCOSE-6-PHOSPHATE DEHYDROGENASE DEFICIENCY

G6PD is a key enzyme that protects erythrocytes from oxidative stresses. Deficiency caused by mutations in the *G6PD* gene causes intravascular hemolysis.¹⁰⁴ The gene is located on the X chromosome, and hence males carrying the affected gene have more severe hemolysis. The severity also depends on the nature of the genetic defect. The G6PD variant (Mediterranean) in parts of India and Pakistan leads to hemolysis only in response to oxidative stress.¹⁰⁵ Individuals deficient in the enzyme cannot maintain an adequate level of reduced glutathione, leading to the precipitation of oxidized hemoglobin in red blood cells, which are then sequestered and lysed. Hemolytic crisis develops within hours of exposure to the stress, usually in the form of drugs, toxins, or infections. Specific causes include pharmacologic agents such as primaquine, sulfonamides, acetylsalicylic acid, nitrofurantoin, nalidixic acid, furazolidone, niridazole, doxorubicin, and phenazopyridine, toxic compounds such as naphthalene balls, infections such as viral hepatitis, rickettsiosis, typhoid fever, and urinary tract infections, and severe metabolic acidosis of any cause. Passage of dark urine followed by oliguria is the most common presentation.¹⁰⁶

Haemolysis resulting from G6PD deficiency may contribute to 5% to 10% of AKI, precipitated by multiple medications and/or infections.⁶ Estimation of the G6PD level in the erythrocytes by the fluorescent spot test confirms the deficiency. Normally, the enzyme activity decreases as the cells age, and older cells with the lowest enzyme activity are destroyed first in a crisis. This process can result in a false-negative test result during a hemolytic episode when the surviving red blood cell population consists of younger erythrocytes, especially in an individual with mild deficiency. The test should, therefore, be repeated after the patient has recovered from the acute episode to confirm the diagnosis. In practice, patients are routinely screened for G6PD deficiency before prescribing a drug that is known to cause hemolysis in such patients, at least in larger centers. Prevalence has been reported to be 8.5% and 7.7% in Indian and selected tribal populations, respectively.^{107,108}

ACUTE CORTICAL NECROSIS

Acute renal cortical necrosis is the most catastrophic of all types of AKI. Of more than 2900 patients with AKI treated with dialysis over 28 years in a study in northern India, 3.8% were found to have acute cortical necrosis.¹⁰⁹ It can develop after a variety of conditions, with the most common being obstetric complications and snake bite. Obstetric complications were responsible in 56% of all cases of acute cortical necrosis, whereas snake bite accounted for 14%. In children,

the most common cause is hemolytic-uremic syndrome. The most striking feature of this condition is prolonged oliguria or anuria. This phase may extend for weeks to months, and patients with diffuse cortical necrosis may never enter a diuretic phase.^{110,111}

Renal recovery depends on the amount of viable cortical tissue and can be slow and incomplete as the surviving nephrons hypertrophy to compensate for the lost nephron mass. In the study in northern India, only 17% of patients could discontinue dialysis by the end of 3 months.¹⁰⁹ Kidney function deteriorates with time in patients who have achieved partial functional recovery. The longest recorded dialysis-free survival has been 12 years.¹⁰⁹

The gold standard for establishing the diagnosis is renal biopsy, although CT has emerged as a reliable noninvasive imaging modality for the early diagnosis of acute cortical necrosis.^{112,113} The characteristic findings include a lack of enhancement of the renal cortex, except for the subcapsular rim after injection of a contrast agent, medullary enhancement, and absence of pelvicalyceal excretion.¹⁰⁹ Cortical tram-track or eggshell calcification develops later and may be detected on a plain radiograph, ultrasonogram, or CT scan. Histologic examination shows a variable degree of necrosis of all elements of the renal parenchyma, especially the cortical region.¹⁰⁹ Some cortical tissue in the subcapsular and juxtamedullary regions may be spared, and its hypertrophy is responsible for the partial recovery of renal function. Other findings include fibrin thrombi in the glomerular capillaries, fibrinoid necrosis of vessel walls, calcification of the necrotic areas, and cortical hemorrhages. The lesions may be classified into patchy and diffuse types, depending on whether the entire parenchyma or only a part of the renal tissue examined shows features of acute cortical necrosis.¹⁰⁹ Needle biopsy can at best give only an approximate idea of the extent of the lesions and can underestimate or overestimate the extent of lesions because of sampling error.

The pathogenesis of acute cortical necrosis remains unclear. The main hypotheses are vasospasm of small vessels and toxic capillary endothelial damage.¹⁰⁹ Prolonged vasospasm of both cortical and medullary vessels induces cortical necrosis in experimental animals. The reasons for increased propensity for renal cortical necrosis during pregnancy are not clear. Renal vasculature in pregnancy may be more prone to vasoconstriction secondary to the effect of sex hormones. Similarities between acute cortical necrosis and the generalized Schwartzman reaction induced in experimental animals by the injection of endotoxin have also been noted.¹¹⁴ Unlike in nonpregnant animals, in which two small doses of endotoxin administered 24 hours apart cause this phenomenon, only one injection is sufficient in pregnant rabbits.¹¹⁵

The presence of fibrin thrombi in the vasculature of patients with acute cortical necrosis has led to consideration of intravascular coagulation as the initial event. A role for endothelium-derived vasoactive substances in the genesis of acute cortical necrosis has also been proposed.¹¹⁰ The increased endothelin-1 levels in women with preeclampsia could contribute to renal ischemia, and a potential role for polymorphisms in the endothelin-1 gene has been suggested.¹¹⁶ However, more studies are needed to establish the exact role of endothelin in the pathogenesis of acute cortical necrosis.

ACUTE PERITONEAL DIALYSIS FOR ACUTE KIDNEY INJURY

As with all AKI, supportive care and removal or reversal of the cause are the mainstay of treatment; however, some patients will recover kidney function in time to avoid dialysis. Where resources are scarce, acute dialysis may not be available, and these patients will die.

PD becomes important in such resource-constrained settings because it requires minimal infrastructure and expertise. Whether peritoneal dialysis is similar or inferior to hemodialysis has been the subject of a Cochrane review protocol, but appears to be noninferior.^{117,118}

However, when required, and if available, PD is likely to be lifesaving. There have been several success stories of acute PD being lifesaving in desperate and emergency circumstances.¹¹⁹ Rigid temporary catheters, which are inserted over a stylet at the bedside, and PD fluid are easily available and inexpensive. Performance of acute PD is a skill that is easily taught and acquired. In fact, many general physicians, especially in the armed forces, who might be posted to far-off places or places where hemodialysis is not available around the clock, routinely carry out acute PD whenever required. Because facilities may be inadequate and overloaded, and delay is inevitable, the timing of dialysis is also an issue. A single-center, open-label, randomized controlled trial has investigated the early initiation of dialysis versus traditional indication-based initiation of dialysis in 208 subjects with community-acquired AKI in India. The investigators did not find any differences in mortality or dialysis dependence at 3 months between the two groups.¹²⁰

CHRONIC KIDNEY DISEASE

INCIDENCE AND PREVALENCE OF CHRONIC KIDNEY DISEASE

An accurate estimate of the number of patients on the Indian subcontinent who have CKD or need RRT is not available owing to the lack of nationwide registry data.¹²¹ A countrywide cohort study of early-stage CKD patients has been started in India. Over the next 5 years, approximately 5000 patients would be enrolled and followed up for the next 5 years. The study aims to describe the epidemiology of CKD progression and the development of complications such as cardiovascular disease in CKD in developing countries. In India, Mani has reported a prevalence of CKD of 1.1% among a rural population of 25,000 who were subjects of a universal screening program in which the serum creatinine concentration was measured only in those with hypertension or proteinuria.¹²² Agarwal and colleagues screened 4700 adults in an urban community and found a point prevalence of 7852/million individuals with a serum creatinine level higher than 1.8 mg/dL.¹²³

Another cross-sectional study that screened 5588 adults from different parts of India reported a CKD prevalence of 17.2%, with about 6% having CKD stage 3 or worse.¹²⁴ The prevalence of early stages of CKD (stages 1–3) was reported to be about 13.1% to 15% using the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) or Modification of Diet in Renal Disease study (MDRD) equation in a cohort

of 3398 otherwise healthy adults.¹²⁵ Studies from Pakistan,¹²⁶ Nepal,¹²⁷ and Bangladesh¹²⁸ have reported a CKD prevalences of 25.3%, 10.7%, and 13.1% to 16%, respectively, in the general population. These figures must be interpreted with caution because of the wide variations in the definition of CKD, study methodology, and sampled population. Despite a high prevalence of CKD, there is a low awareness of the disease; only 7.9% and 2.3% of subjects were aware of the disease in the studies from India¹²⁴ and Pakistan,¹²⁶ respectively. Lack of both health care resources and education is possibly responsible for the low awareness of disease.

Commonly used formulas for the estimation of the glomerular filtration rate (GFR), such as the MDRD equation, have been validated only in small population-based studies in the Indian subcontinent.^{129,130} Differences in body habitus and dietary habits make it likely that these formulas may require further validation and possibly correction factors for the accurate assessment of GFR in this population.¹³⁰ Two of the later studies showed the mean measured GFR in healthy Indian adults (kidney donors) to be only 81 mL/min, substantially lower than values reported in the West.^{131,132} This finding raises the question of whether the thresholds used to define CKD should be modified. The crude and age-adjusted incidences of ESKD in India have been estimated to be 151 and 232/million population, respectively.¹³³ This means that 250,000 to 300,000 new patients need RRT every year. Data on the prevalence of ESKD are not available.

DEMOGRAPHICS OF CHRONIC KIDNEY DISEASE AND END-STAGE KIDNEY DISEASE

In the past, glomerulonephritis was reported to be the most common cause of ESKD in the Indian subcontinent.¹³⁴ The high prevalence of glomerular diseases was linked to the prevalence of bacterial and viral infections. The causative spectrum, however, has been changing in the past decade (Table 79.3).

Diabetic nephropathy, previously restricted to high-income urban residents and older individuals, has now emerged as the most important cause of CKD in this region (see discussion

Table 79.3 Causes of Chronic Kidney Disease and End-Stage Kidney Disease in India and Pakistan (%)

Cause	India (N = 57,273) ^a	Pakistan (N = 6127) ^b
Diabetic nephropathy	31.3	40.35
Hypertension	12.9	28.89
Chronic glomerulonephritis	13.8	11.57
Chronic interstitial nephritis, calculus disease, obstructive nephropathy	10.4	7.6
Other, unknown	31.8	11.48

^aAdapted from Rajapurkar MM, John GT, Kirpalani AL, et al.

What do we know about chronic kidney disease in India: first report of the Indian CKD registry. *BMC Nephrol.* 2012;13:10.

^bData from The Kidney Foundation. Dialysis Registry of Pakistan 2007–2008. Karachi, Pakistan: The Kidney Foundation; 2008.

on diabetic nephropathy in Chapter 39). According to the Indian CKD Registry, which has information on more than 57,000 patients, diabetes was listed as the primary diagnosis in 31% of cases of the CKD.¹³⁵ The frequency increased to 40% in incident ESKD cases.¹³³ In older adults (>60 years), diabetic nephropathy accounted for about 58% of all cases of CKD¹³⁶; however, this is not a nationwide registry. According to the Dialysis Registry of Pakistan, more than 40% of cases of ESKD are due to diabetic nephropathy.¹³⁷ In Sri Lanka and Bangladesh also, diabetic nephropathy is an important cause of CKD.^{138,139} This change has paralleled the increase in the prevalence of type 2 diabetes in the general population, especially in the areas undergoing rapid urbanization.¹⁴⁰ The projected increase in the prevalence of diabetes in the region suggests that these numbers will rise further over the next 15 years.¹⁴¹

The role of the intrauterine environment in the development of chronic disease in adults, particularly systemic arterial hypertension and CKD, has come to the fore and could explain the link between poverty and malnutrition in the mother and the subsequent development of CKD in the offspring. Low birth weight and early malnutrition, followed by overnutrition in adult life, have been shown to be associated with the development of metabolic syndrome, diabetes, and diabetic nephropathy in an Indian cohort.¹⁴² Whether nephrogenesis is influenced by intrauterine malnutrition and/or any adverse intrauterine environment is a matter of ongoing investigation (see discussion of nephron endowment and developmental programming of blood pressure and renal function in Chapter 21). The finding of a high prevalence of proteinuria and high blood pressure in southern Asian children could be part of this jigsaw puzzle.^{143,144} Also not investigated has been the role of dietary habits and indigenous medicines. Whether any of these factors has an adverse effect on kidney function remains unknown. Infections such as chronic hepatitis B or C, HIV, and tropical infectious diseases are prevalent in this part of the world, and renal involvement is not uncommon in these infections. Although there are no credible data that support these infections as a significant cause of CKD, this possibility cannot be ruled out because it is likely that this is missed in practice and hence not recorded.¹⁴⁵

The mean age of patients with CKD, including those requiring RRT, is generally lower in this region than in other parts of the world. The mean age of Indian patients with CKD is 50.1 ± 14.6 years; 70.3% are males and 29.7% females, according to the Indian CKD Registry.¹³⁵ This lower mean age is likely related to unique environmental exposures at a younger age and the poor availability of health care, which delays diagnosis and leads to a loss of opportunities to institute timely preventive measures, culminating in faster progression to ESKD. Dialyzer reuse offers cost savings and is common.¹⁴⁶ Social and infrastructural limitations lead to poor compliance, even when dialysis is offered free of cost.¹⁴⁷ Certain causes of CKD have been shown to predominate in specific geographic locations, as discussed later.

CHRONIC KIDNEY DISEASE DUE TO GLOMERULONEPHRITIS

In the absence of a biopsy registry, the exact spectrum of glomerulonephritis cannot be known. However, a large study of more than 5400 kidney biopsy specimens at a south Indian

tertiary care center that treats not only patients from India but also those from neighboring countries has provided an insight into the range of glomerulonephritis in the region.¹⁴⁸ Primary glomerulonephritis was diagnosed in 71% of all the biopsy specimens. Mesangioproliferative glomerulonephritis without IgA was the most common lesion (20.2%), followed by idiopathic focal segmental glomerulosclerosis (FSGS) (17%), minimal change disease (11.6%), membranous nephropathy (9.8%), IgA nephropathy (8.6%), and membranoproliferative glomerulonephritis (3.7%). Postinfectious glomerulonephritis accounted for 12.3% of all lesions.

To ascertain changing trends, biopsy data collected between 1971 and 1985 were compared with later data in the same study. The later data showed significant increases in FSGS (17% vs. 8.6%) and membranous glomerulonephritis (9.8% vs. 6.4%) during the period 1986 to 2002, whereas minimal change disease (11.6% vs. 16.5%) and membranoproliferative glomerulonephritis (3.7% vs. 7.2%) decreased significantly during the same period. Thus, in comparison with other registries, the prevalence of IgA nephropathy is much lower than that in East Asians, whereas FSGS is more common. Similar trends have been reported in another study from the northern region of the subcontinent.¹⁴⁹ The analysis of data for adults presenting with nephrotic syndrome due to primary glomerular disease has revealed that FSGS is the most common lesion in both India and Pakistan. The frequency of occurrence of other primary glomerular diseases in these patients is presented in Table 79.4.¹⁴⁹⁻¹⁵¹ The morphologic categories of FSGS in the Indian population have been described and are different from those in the West.¹⁵² There is a lower prevalence of the perihilar and collapsing variants and a higher prevalence of tip and cellular variants. Secondary amyloidosis is an important cause of glomerular disease on the subcontinent and is much more common than primary amyloidosis. Tuberculosis is the main cause of secondary amyloidosis in India, accounting for two-thirds of cases, whereas rheumatoid arthritis is responsible in only 6% of cases.^{153,154}

CHRONIC KIDNEY DISEASE DUE TO VASCULAR DISORDERS

The most common causes of renovascular hypertension worldwide are fibromuscular dysplasia in the young and atherosclerosis in older adults. On the Indian subcontinent, however, Takayasu arteritis or nonspecific aortoarteritis is the main cause of renovascular hypertension in young adults, as well as in children, accounting for 59% to 80% of all cases in these groups.^{155,156} In older adults, as elsewhere, atherosclerosis is the most common cause.

Takayasu arteritis is an inflammatory vascular disease of unknown cause predominantly affecting young females in their second and third decades. There is a potential association described with tuberculosis.¹⁵⁷ Takayasu arteritis involves the large elastic arteries and results in occlusive or ectatic changes, mainly in the aorta and its major branches. The average age of Indian patients is between 25 and 30 years.¹⁵⁸

Aortoarteritis has been classified into the following types, according to the site of involvement^{159,160}:

- Type I affects the branches of the aortic arch.
- Type IIa affects the ascending aorta, aortic arch, and its branches.

Table 79.4 Spectrum of Primary Glomerular Lesions Causing Nephrotic Syndrome on the Indian Subcontinent^a

Disorder	India (N = 324) ^b	Pakistan (N = 316) ^c	Nepal (N = 137) ^d
Focal segmental glomerulosclerosis	99 (30.6)	126 (39.9)	11 (8.0)
Membranous nephropathy	79 (24.4)	84 (26.6)	58 (42.3)
Membranoproliferative glomerulonephritis	58 (17.9)	14 (4.4)	30 (21.9)
Minimal change disease	48 (14.8)	50 (15.8)	14 (10.2)
Postinfectious glomerulonephritis	9 (2.8)	9 (2.8)	4 (2.9)
Immunoglobulin A nephropathy	6 (1.8)	8 (2.5)	3 (2.2)
Other	25 (7.7)	25 (8.0)	17 (12.4)

^aNumbers in parentheses indicate percentages.

^bData from Rathil M, Bhagat RL, Mukhopadhyay P, et al. Changing histologic spectrum of adult nephritic syndrome over five decades in Northern India: a single center experience. *Ind J Nephrol*. 2014;24:13–18.

^cData from Kazi JI, Mubarak M, Ahmed E, et al. Spectrum of glomerulonephritides in adults with nephrotic syndrome in Pakistan. *Clin Exp Nephrol*. 2009;13:38–43.

^dData from Gargal, Kafle RK. Histopathological spectrum of glomerular disease in Nepal: a seven-year retrospective study. *Nepal Med Coll J*. 2008;10:126–128.

- Type IIb affects the ascending aorta, aortic arch, and its branches and the descending thoracic aorta.
- Type III affects the thoracic aorta, abdominal aorta, and/or renal arteries.
- Type IV affects the abdominal aorta and/or renal arteries.
- Type V is a combination of types IIb and IV.

Involvement of a coronary or pulmonary artery is indicated by appending the suffix *C(+)* or *P(+)* to any of the types.¹⁶¹ Involvement of the abdominal aorta and/or renal arteries is more common in India and other Southeast Asian countries and South America than in Japan, where the aortic arch and its branches are more commonly involved. In Indian and Bangladeshi patients, type III disease is the most common, accounting for 53% to 76% of all cases.¹⁶²

In a large study of 650 cases in India, clinical manifestations of aortoarteritis included unequal pulses (96%), hypertension (72%), oliguria due to renal failure (30%), intermittent claudication (25%), neurologic symptoms (e.g., amaurosis fugax, syncope, transient ischemic attacks; 22.5%), eye changes (8.1%), and skin manifestations (e.g., erythema nodosum, Raynaud phenomenon, leg ulcers; 3.8%).¹⁶² It is important to record blood pressure in all four limbs because arteries supplying the upper limbs are often involved in the disease process.

The renal artery is narrowed at its ostium and in the proximal third. Bilateral renal artery stenosis is seen in about 50% of all patients (Fig. 79.2). Histologic findings in affected vessels vary according to the stage of the disease. In the early stages, granulomatous inflammation and infiltration with polymorphs, mononuclear cells, and multinucleated giant cells are seen in all the layers but are more marked in the adventitia than in the media or intima.¹⁶³ These features may also be seen around the vasa vasorum, which show endothelial proliferation and obliteration of the vessel lumen. In more advanced disease, the inflammatory process is less evident, but adventitial fibrosis and intimal smooth muscle proliferation and fibrosis result in marked luminal narrowing.

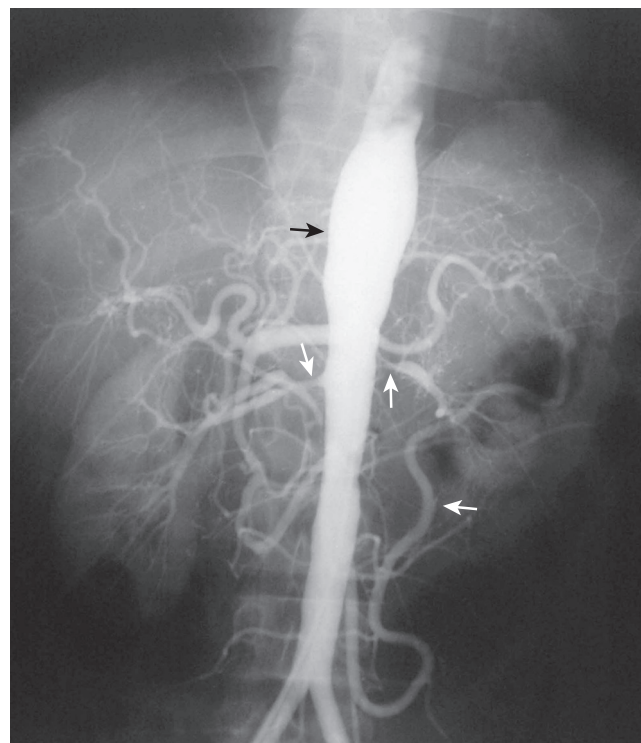


Fig. 79.2 Takayasu arteritis. Shown is bilateral renal artery narrowing (vertical arrows) with poststenotic dilation on the left side, dilation and irregularity of the abdominal aorta (black arrow), and a large collateral vessel (horizontal arrow) arising from the inferior mesenteric artery.

Nonspecific ischemic glomerular lesions resulting from arterial narrowing and hypertension are frequently observed in patients with renal artery involvement. Rarely, glomerular lesions, such as mesangioproliferative, focal proliferative, membranoproliferative, and crescentic forms of glomerulonephritis, have been reported. Renal amyloidosis in association with Takayasu arteritis has also been reported in rare cases.¹⁶⁴

Prednisolone, at a dosage of 1 mg/kg body weight per day, tapered to 15 mg/day by 3 months,¹⁶⁵ often dramatically improves the constitutional symptoms, halts disease progression in patients with inflammatory stage disease, and lowers the erythrocyte sedimentation rate (ESR) toward normal. However, even the ESR is not a reliable marker of disease activity. If progression of disease is seen in patients undergoing steroid therapy, cytotoxic drugs such as cyclophosphamide or azathioprine may be used.¹⁶⁶ Alternatively, low-dose methotrexate may enhance the efficacy of steroid therapy and facilitate steroid sparing.¹⁶⁶ The results of balloon angioplasty with stent placement in narrowed arteries are highly encouraging.¹⁶⁶ In patients in whom stents cannot be implanted, surgical reconstruction may be carried out after the disease becomes inactive.

CHRONIC KIDNEY DISEASE DUE TO RENAL CALCULI

Obstructive nephropathy due to urolithiasis is common in Pakistan and contiguous parts of northern India. This constitutes what has been referred to as a “renal stone belt,”¹⁶⁷ where nephrolithiasis is responsible for 5% of all cases of ESKD.¹⁶⁷

The available literature on urinary calculi shows a different stone pattern across the world, highlighting different geographic and causative factors. Studies in northern India have shown that the vast majority of calculi (>90%) are calcium oxalate stones, predominantly calcium oxalate monohydrate (80%).¹⁶⁸ Apatite, struvite, and uric acid stones account for less than 2% each. This incidence of calcium oxalate monohydrate stones is significantly higher than that in Western countries, where such stones constitute up to 55% of the total.¹⁶⁹ Typically, staghorn calculi are composed of calcium magnesium ammonium phosphate. However, in Indian studies, even staghorn calculi are found to be made up predominantly of calcium oxalate. Reasons for the high incidence of calcium oxalate calculi in this region could include the oxalate-rich vegetarian diet, along with its high carbohydrate content and the high mineral and fluoride content of drinking water. Fluoride is thought to promote stone formation by increasing oxaluria and insoluble calcium fluoride in urine.¹⁷⁰

CHRONIC KIDNEY DISEASE DUE TO INDIGENOUS THERAPIES

HERBAL MEDICINE TOXICITY

On the subcontinent, a combination of ignorance, poverty, nonavailability of health care facilities, high cost of modern medicines, and the widespread belief in indigenous systems drives people to turn to indigenous drugs. It is commonly believed that these remedies are gentler and have no side effects. Herbal remedies are often classified as dietary supplements for regulatory and marketing purposes and hence are exempt from rigorous safety testing. However, adulteration of herbal medicines is common in many countries. A state government report has cited finding undeclared pharmaceuticals or heavy metals in 32% of Asian patent medicines sold in California.¹⁷¹ The high heavy metal content of indigenous herbal drugs could be due to heavily polluted soil or irrigation water. Dwivedi and Dey have found high

lead and cadmium levels in the leaves of medicinal plants from India.¹⁷²

The medical community has increasingly recognized the potential role of these remedies in causing harm to various organ systems, including the kidneys. Indigenous therapies may cause AKI as well as CKD. In AKI, the cause-and-effect relationship is easier to establish because there is a temporal relation between the intake of the agents and the injury. However, physicians often do not seek the history of intake of natural medicines. Moreover, a chemical analysis of such drugs is seldom carried out. Prakash and colleagues have described the case of a 60-year-old man who had unexplained kidney failure.¹⁷³ Investigation revealed that he had been taking an herbal medicine from India containing a large amount of lead. After discontinuation of the herbal medicine and several sessions of lead chelation therapy, his creatinine level declined. Lead poisoning should be considered as a consequence of traditional remedy-induced kidney injury.¹⁷⁴ As of now, supporting alternative drugs without any scientific evidence is not recommended. However, trying to find reasons for their use in community and discovering their beneficial effects, if any, could be an area for research.

TOXICITY OF NATURAL MEDICINES FROM ANIMAL SOURCES

The raw gallbladder or bile of freshwater and grass carp is used in parts of eastern India to reduce fever, treat cough, decrease hypertension, improve visual acuity, treat rheumatism, and promote general health.¹⁷⁵ A syndrome of acute hepatic and renal failure has been reported in exposed patients.¹⁷⁵ Symptoms appear minutes to hours after ingestion and include abdominal pain, nausea, vomiting, and watery diarrhea. Hepatocellular jaundice is observed in more than 60% of patients. AKI sets in within 48 hours and is oliguric in most patients, and more than 75% also show microscopic hematuria. The duration of AKI ranges from 2 to 3 weeks. The variation in symptomatology is likely to be related to differences in the varieties of fish or amount of bile ingested, as well as in individual susceptibility. Renal histologic examination reveals tubular necrosis and interstitial edema.

The mechanism whereby AKI develops is not well understood and may include bradycardia and hypotension owing to the cardiotoxic effect of the bile salts.¹⁷⁵ Bile salts also inhibit intestinal Na⁺-K⁺-ATPase, which increases mucosal permeability and leads to diarrhea.¹⁷⁵ Bile produces diuresis, excessive salt loss, and cardiac depression in rats.¹⁷⁶ Hypotension and hemolysis may also contribute to renal failure. Recovery has been universal among patients who have sought medical attention in a timely manner, and death has occurred only in those who reported late and had multiorgan failure.

CHRONIC KIDNEY DISEASE OF UNCERTAIN CAUSE

A large proportion of Indian patients with ESKD come for treatment with a relatively short history of kidney problems and advanced renal failure with no previous health records.¹⁶⁷ This makes the task of determining the primary disease difficult. Thus there is a large proportion of patients with ESKD of unknown cause (CKDu), which accounts for over one-third of all patients at our center. Reports of CKD of CKDu have

also emerged from the north central provinces of Sri Lanka. A detailed epidemiologic investigation has suggested that potential exposure to cadmium through local food chains might be associated with CKDu. The mean concentration of cadmium in urine was found to be significantly higher in those with CKDu as compared with controls in the endemic and nonendemic areas ($1.039 \mu\text{g/g}$ vs. $0.646 \mu\text{g/g}$, $P < .001$; $1.039 \mu\text{g/g}$ vs. $0.345 \mu\text{g/g}$, $P < .05$, respectively).¹⁷⁷ Also, dietary selenium deficiency and genetic factors have also been proposed as possible risk factors for development of CKDu.^{177,178} In a recent systematic review, positive family history, male gender, farming occupation, middle age, snake bite, and possible heavy metal exposure were found to be associated with CKDu.¹⁷⁹ Most of the affected individuals were male paddy farmers of low socioeconomic status who had progressive nonproteinuric CKD.^{180,181} Biopsy specimens showed tubulointerstitial nephritis, with minimal inflammation and extensive fibrosis. The disease bears a strong resemblance to Balkan nephropathy and Chinese herbal nephropathy. CKDu is discussed further in Chapter 76.

END-STAGE KIDNEY DISEASE

FINANCIAL AND REIMBURSEMENT ISSUES

Unlike in Western nations, the concept of health insurance (both government-funded and private) is still in a primitive stage on the Indian subcontinent. The costs of RRT, therefore, have to be borne by most patients out of their own funds.¹⁸² Some government and private organizations cover the cost of treatment of employees and their dependents as part of employment benefits. The overall cost of RRT is less in dollar terms than in the industrialized countries because of the lower staff salaries and the low cost of drugs. Nevertheless, it is still several times higher than the per capita gross national product and remains out of reach for most of the population.

One Indian study has reported that about two-thirds of patients who needed RRT obtained financial assistance from employers or accepted charity, one-third sold property or family valuables, and one-quarter took out loans to cover the cost.¹²² Many patients raised funds in more than one way. Only 4% were able to cover the cost using their family resources.

According to the Indian CKD Registry, about 57% of patients with ESKD are not receiving any replacement therapy.¹³⁵ Among those who started dialysis in one unit in India, 28% were able to sustain treatment beyond 3 months.¹⁸³ Among older adults, one study has reported that of those who started HD, only 18% were still receiving dialytic support at the end of 1 year.¹³⁶ Similar high dropout rates (18%) were observed at the Sindh Institute of Urology and Transplantation, which provides free dialysis in Pakistan,¹⁴⁷ illustrating that the provision of free dialysis is not enough to overcome the access barriers in the region. The cost of dialysis is also influenced by late presentation, with resulting poor clinical status that necessitates hospitalization. Poor hygiene, hot and humid climate, and overcrowding predispose to a variety of life-threatening infections. It has been reported that 12% to 18% of all patients undergoing dialysis eventually have tuberculosis.¹⁸⁴

HEMODIALYSIS

There are about 5500 dialysis centers in India, with over 90% in the private sector.¹⁸⁵ Pakistan had 140 dialysis centers in

2004, which increased to 175 in 2009.¹⁸⁵ They are spread over 53 cities—about 30% are government-funded, and 45% are under private management. The rest are run by community support or charitable agencies. In both countries, a large number of dialysis units are small minimal care facilities, owned and looked after by non-nephrologists or even technicians.¹⁸⁵

Decisions on the frequency and duration of HD sessions are based on patient symptomatology, financial considerations, and the availability of dialysis slots. Most patients undergo one or two 4-hour sessions every week. Dialyzer reuse is practically universal, and reprocessing is often performed manually. The absence of regulation by the government or professional societies has prevented standardization of dialysis procedures, including the establishment of minimum standards for dialysis machines, water quality, type of dialyzers, and reuse policies.¹⁸²

Viral hepatitis is among the most common viral infections encountered in patients undergoing dialysis. Hepatitis B vaccination, despite low seroconversion rates, has reduced the prevalence from 32% to 4.7% among patients in India.¹⁸⁶ Hepatitis C virus (HCV) has emerged as the predominant cause of viral hepatitis in patients undergoing maintenance HD. The annual incidence of HCV infection as detected by anti-HCV antibodies among patients at an Indian HD center was reported to be as high as 18%, compared with about 2% in patients receiving continuous ambulatory PD.¹⁸⁷ The prevalence of HCV was found to be 27.7% when HCV RNA was tested in patients receiving HD at a tertiary care hospital.¹⁸⁸ The high incidence of HCV in HD units could be related to the high prevalence of HCV seropositivity in patients undergoing HD, total transfused blood volume, lack of enforcement of universal precautions, high comorbid illness burden, and greater frequency of intervention. It has been shown that the isolation of HCV-infected patients during HD significantly decreases the HCV seroconversion rate, from 36.2% to 2.7%.¹⁸⁹

Malnutrition affects most patients receiving dialysis in this region; the reported frequency is 44% to 77%.¹⁹⁰ The imposition of protein restriction in patients who are already consuming a calorie-deficient diet, delay in initiation of dialysis, and delivery of inadequate dialysis contribute to this problem. In fact, in one study, protein malnutrition was found to increase in as many as 86% of Indian patients after the initiation of dialysis.¹⁹¹

PERITONEAL DIALYSIS

Facilities for HD are available only in the larger cities. Because PD can be done at home, it could be regarded as the preferred form of dialysis for most patients who live in rural areas and small towns. Despite this advantage, PD continues to be grossly underused. The major reason is its higher cost in comparison with HD on the subcontinent. Other reasons include delayed presentation to dialysis units, which gives insufficient time for patient education and preparation, and the fact that many nephrologists are not adequately trained to provide PD. Concerns are often raised on the grounds that poorly educated patients are likely to be nonadherent to therapy and would be at greater risk of peritonitis owing to the hot humid climate and poor hygienic conditions.

Although PD has been available for almost 25 years in India, at our center this modality is used in fewer than 20%

of patients. In 2008, the number of patients treated with long-term PD in India and Pakistan was only about 6500 and 100, respectively.¹⁹² Patients are assigned to PD not as a matter of choice but mainly because they have contraindications for other modalities of RRT. In one large hospital, only 8% of PD recipients were started on PD directly; 92% were shifted after receiving HD for a mean duration of about 6 months. Of those, two-thirds were switched because they tolerated HD poorly, 30% because of comorbid conditions and vascular access problems, and 3% because of lifestyle issues.¹⁹³ Patients who initially underwent PD were more likely to have diabetes mellitus and coronary artery disease.¹⁹³ Their average age was 63 years, compared with 34 years for the HD population.

The patient's economic status also determines the PD prescription. The most common practice is to start patients on three 2-L exchanges daily.¹⁹⁴ It is not uncommon for patients to reduce the number of exchanges as their financial resources dwindle and eventually die of complications of underdialysis. Cycler-assisted PD and the use of newer solutions remain the exclusive privilege of the rich.

In addition to factors directly related to PD, cardiovascular disease is an important killer. Patients often do not undergo cardiovascular evaluation, having already exhausted their resources on PD. This scenario is similar to that seen in the early years of PD practice in Mexico, where aggressive marketing led non-nephrologists to start PD without attention to quality.¹⁹⁵

Peritonitis remains the major problem associated with PD on the Indian subcontinent. The initial rate of peritonitis was one episode every 5 to 6 patient-months,¹⁹⁶ but this has declined significantly as training has improved and patients have been switched to the double-bag system. Some programs reported a higher incidence of infections in the summer months.¹⁹⁶ Peritonitis rates are now around 0.39 to 0.41 episodes/patient-year.¹⁹⁷⁻²⁰¹ Two unique features of PD-related infections in Indian patients are a high rate of culture-negative cases and a predominance of gram-negative peritonitis, mainly in northern India. Culture-positive peritonitis was found in only 63% to 72% of episodes.^{197,198}

Gram-negative bacteria constitute 60% to 66% of all organisms found on culture of patients undergoing PD.²⁰²⁻²⁰⁵ *E. coli* is usually found. Other organisms are *Klebsiella pneumoniae*, *Acinetobacter calcoaceticus*, *Pseudomonas aeruginosa*, and *Enterobacter* spp. Overall, organisms of fecal origin are more common than those of skin origin. This finding could be related to the unique habit of ablution after defecation in the region, which facilitates the transfer of fecal organisms to the hand. Patients with gram-negative peritonitis require more frequent hospitalization and have higher rates of catheter loss, switch to HD, and death.¹⁹⁸

In southern India, Pakistan, and Nepal, gram-positive infections are predominant, in contrast to the pattern of infections seen in North India.^{202,205} Fungal peritonitis is noted in about 10% to 14% of all peritonitis episodes.²⁰³ More than 90% of these are *Candida* infections. Most episodes are preceded by bacterial peritonitis. Prompt catheter removal improves outcomes. Despite the high frequency of mycobacterial infections in the community, tuberculous peritonitis has not emerged as a significant problem in the Indian PD population.

Malnutrition is a major problem in these patients. Prasad and associates have shown that nutritional status at the time of PD initiation was predictive of peritonitis rates. Malnourished

patients experienced significantly more peritonitis episodes than patients with normal nutritional status (1.0 vs. 0.2 annually).²⁰⁶

KIDNEY TRANSPLANTATION

Constraints in operating an effective maintenance dialysis program leave renal transplantation as the only viable option for patients with ESKD. However, transplantation activity falls woefully short of demand; lack of finances, lack of an organized cadaver-donor transplant program, and social issues are the major stumbling blocks. Despite these limitations, community participation coupled with strong public and private sector efforts can establish a free and self-sustaining RRT program, as has been demonstrated in Karachi, Pakistan; 1- and 5-year graft survival rates of 92% and 85%, respectively, have been reported, which are highly encouraging.²⁰⁷

Cadaver donors are seldom used because of absence or ineffectiveness of an organ procurement network, lack of facilities for taking care of potential donors, and poor public education. The process depends on the initiative of individual transplant physicians, surgeons, and cooperating ICUs. Even though more than 70,000 road fatalities were recorded annually in India in 2014,²⁰⁸ lack of prompt transport and unavailability of life support services preclude organ donation, even in situations in which the families could be approached for consent.²⁰⁹ Of around 4000 renal transplantations performed annually in India, only 2% use organs from cadaveric donors. For transplantations involving living related donors in India, the proportion of spousal donors (mainly wives) has increased over the past decade; they constitute about 40% of all donors.²¹⁰

Affordability is another major barrier. Even though patients do not have to bear hospitalization costs in state-subsidized hospitals, the cost of immunosuppressive therapy is not reimbursed. In a prospective analysis of 50 kidney transplant recipients in India, direct expenses for kidney transplantation—physician fees, cost of drugs and disposables, dialysis, and costs of laboratory investigations and hospitalization—were estimated to range from \$2151 to \$23,792 and indirect expenses—travel, food, stay, and loss of income—from \$226 to \$15,283 (all in U.S. dollars).²¹¹ Overall, about 54%, 8%, and 10% of families suffered from severe, moderate, and some financial crisis, respectively.²¹¹ Antibody induction therapy and prophylaxis for cytomegalovirus infection are therefore rarely used. Pretransplantation HCV infections also often go untreated. Patients are nonadherent with regimens of expensive drugs such as calcineurin inhibitors, leading to high rates of graft loss. The cost of treating steroid-resistant rejections is prohibitive. Cost reduction strategies that are frequently used include limiting induction therapy to high-risk patients, using cytochrome P450 inhibitors (e.g., ketoconazole, non-dihydropyridine calcium channel blocker), using azathioprine instead of mycophenolate mofetil, continuing prednisolone long term, and using bioequivalent generic drugs.

The worldwide shortage of organs for transplantation has given rise to the practice of the purchase of kidneys from poor donors by affluent persons in India in the 1980s and early 1990s.²¹² The buyers came from both within and outside the country—hence, the term “transplant tourism.” The exploitation of donors and substandard medical care provided

to recipients were widely condemned and prompted the enactment of a law by the Indian Parliament in 1994 officially banning this practice. Since then, it has been carried out only clandestinely in India; it is more common in some parts of Pakistan, although exact numbers are not known.^{213,214} The Human Organ and Tissue Transplantation Ordinance, which was passed in September 2007 in Pakistan, explicitly and unambiguously makes the buying and selling of human organs a crime and prohibits transplantation of organs from Pakistanis into foreigners.²¹⁵ Of late, more transplants from unrelated living donors are being done in Sri Lanka, as reported in the lay press.

Infections complicate the course in 50% to 75% of kidney transplant recipients in the region, with mortality ranging from 20% to 60%.¹⁹⁰ Rubin has categorized these infections as those occurring within the first month after transplantation, those occurring within 2 to 6 months, and those occurring thereafter.²¹⁶ The reported 6-month milestone of a decrease in the susceptibility to infections after a step-down of immunosuppression is not seen in these patients. This reflects an altered susceptibility pattern caused by coexisting infections in immunosuppressed patients in the region, together with a higher prevalence of endemic infections.²¹⁷

Tuberculosis affects 10% to 13% of renal transplant recipients in India²¹⁸⁻²²⁰ and results from the reactivation of a quiescent focus. It manifests in the first year after transplantation in more than 50% of patients. Although pleuropulmonary involvement is the most common, disseminated disease occurs in about 30% of patients. Unusual sites of involvement include the skin, tonsils, vocal cords, and prostate.

Renal transplant recipients with tuberculosis present numerous diagnostic difficulties. The Mantoux test is generally unhelpful, classic radiologic findings are seen only in a minority, examination of a sputum smear for acid-fast bacilli has a low yield, and culture takes 4 to 6 weeks. Bronchoalveolar lavage, bone marrow biopsy, and liver biopsy must be used to make a diagnosis of tuberculosis.

There are also problems with the treatment of tuberculosis in transplant recipients, specifically in selecting antituberculous drugs and determining the duration of therapy. Rifampicin is a well-known hepatic P450 microsomal enzyme inducer that increases the clearance of both prednisolone and calcineurin inhibitors. The dosage of calcineurin inhibitors needs to be increased threefold to fourfold to maintain therapeutic blood levels. This change raises the cost of therapy and is unacceptable to the vast majority of patients. An alternative regimen that has been successfully used consists of a combination of isoniazid, ethambutol, pyrazinamide, and ofloxacin or ciprofloxacin.²²¹ The optimal duration of therapy is a matter of debate. For combinations using rifampicin and isoniazid, 9 months of treatment have been recommended.²²¹ However, the duration is increased to 18 months for patients who are not receiving rifampicin.²¹⁸ Such patients generally receive two drugs, usually a fluoroquinolone and isoniazid for 18 months. The initiation of isoniazid chemoprophylaxis during dialysis prevents the development of tuberculosis after transplantation. However, isoniazid can cause hepatic dysfunction, for which it often must be discontinued.²²² Drug-resistant mycobacterial strains are also of concern, and the incidence of primary isoniazid resistance has been increasing steadily.²²³ The role of isoniazid prophylaxis thus remains controversial.²²⁴

Other common infections in transplant recipients are hepatitis B and hepatitis C, both of which are highly prevalent in patients undergoing dialysis. Hepatitis B is encountered in about 5%, whereas hepatitis C is seen in 15% to 20%. The cost of interferon therapy and indirect expenses (including dialysis) during therapy for HCV infection have been so prohibitive that the majority ($\approx 75\%$) of HCV-positive patients undergo kidney transplantation without receiving anti-HCV treatment.²²⁵ With the availability of new directly acting antiviral drugs, however, the paradigm is rapidly changing. Most patients now receive treatment and enter sustained remission. Although patient and graft survivals at 5 years were reported to be similar in HCV-positive and HCV-negative patients, serious bacterial infections were significantly more common in HCV-positive patients.^{226,227} The rate of cytomegalovirus infection was found to increase from 4% in the years before calcineurin inhibitors were widely used to 17% in the calcineurin inhibitor era in one autopsy series.¹⁸⁴ Primary infection is seldom seen because the vast majority of both donors and recipients are seropositive.¹⁸⁴ Opportunistic fungal infections occur in 4% to 7% of recipients, but these carry a high mortality rate, more than 65%.^{228,229} Malaria and leishmaniasis are also encountered.²³⁰

SUMMARY

The spectrum of both acute and chronic kidney disease on the Indian subcontinent differs significantly from that encountered in the industrialized world and other developing countries. Community-acquired AKI in otherwise healthy individuals continues to be common in the region. Although glomerulonephritis has been the most common cause of ESKD, it is being replaced by diabetic nephropathy. The lack of an adequate number of nephrologists and the high cost of treatment pose the biggest challenges to the management of patients with kidney disease in this region of the world.



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BOARD REVIEW QUESTIONS

1. A 30-year old previously healthy man has come to the emergency room of a community health center with complaints of loose motions, nausea, vomiting, and fever for 3 days and decreased urine output for 2 days. He is dehydrated and in shock. After initial resuscitation with intravenous fluids and other symptomatic treatment, his blood pressure has risen to 120/80 mm Hg. His investigations reveal serum creatinine of 4.6 mg% and serum potassium of 6.9 mEq/L. Even after 6 hours of treatment, he has not produced any urine, and serum potassium is still 6.5 mEq/L. His transportation to a next higher facility will take another 12 hours. Acute peritoneal dialysis over a stiff catheter is possible in the center. Which of the following is most appropriate in these circumstances?
- Shift the patient immediately to a higher center, where facilities for hemodialysis are available.
 - Start acute peritoneal dialysis over a stiff catheter and shift when the patient becomes stable.
 - Continue with aggressive fluid administration and similar management for another 6 hours and decide later.

Answer: b

Rationale: Acute peritoneal dialysis is necessary in this anuric hyperkalemic patient. Transfer if possible should still occur if possible, but waiting 12 hours for initiation of dialysis may be too long.

2. A 26-year-old woman suffered massive postpartum hemorrhage after childbirth and remained in shock for 6 hours. Subsequently, she developed acute kidney injury (AKI), and her serum creatinine increased to 7 mg%. She has been on hemodialysis for the past 3 weeks, and her 24-hour urine

output is 200 mL. Her urinalysis has shown trace albuminuria and occasional red blood cells, without any casts. What is the most appropriate next step in management?

- Perform a renal biopsy.
- Do a contrast-enhanced computed tomography scan of the kidneys.
- Declare her as suffering end-stage kidney disease and plan for renal transplantation.
- Start her on steroids and plasmapheresis.

Answer: b

Rationale: The patient most likely has cortical necrosis. The diagnosis can be made with contrast CT. If the CT does not suggest cortical necrosis a biopsy may be indicated.

3. In which of the following circumstances would you make a diagnosis of chronic kidney disease (CKD) of undetermined cause?
- CKD with unilaterally small kidney and hypertension, without evidence of renovascular insufficiency
 - CKD in a patient with diabetes of long duration in whom a renal biopsy has not been done to confirm diagnosis
 - CKD with symmetrically small kidneys and no albuminuria, without any risk factors
 - CKD in a patient with a preceding history of edema, nephrotic-range albuminuria, and hypertension for 3 years but no renal biopsy diagnosis

Answer: c

Rationale: The pathophysiology of CKDu is generally a tubulo-interstitial process leading to small kidneys and is associated with absence of proteinuria in the absence of other common risk factors.